Table of Contents: Ideas, Uses, and Introduces

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By lecture (file order). For each **idea** we list **uses** (concepts/groups referenced) and **introduces** (concepts first introduced in that idea block).

Day 1 — Intro to Linear Systems (`day_1.qmd`)

Idea	Uses	Introduces
`linear_systems_what_and_why`	linear_system, linear_systems_core	linear_system
`examples_of_linear_systems`	linear_system, linear_systems_core	—
`from_equations_to_augmented_matrix`	augmented_matrix, elimination_and_rref, linear_system, linear_systems_core	augmented_matrix
`elementary_operations_and_rref`	augmented_matrix, elimination_and_rref, elementary_row_operations, linear_system, reduced_row_echelon_form	elementary_row_operations, reduced_row_echelon_form

Idea	Uses	Introduces
`gauss_jordan_algorithm`	augmented_matrix, elimination_and_rref, elementary_row_operations, gauss_jordan_elimination, pivot, pivoting, reduced_row_echelon_form	gauss_jordan_elimination, pivoting, pivot
`rounding_mining_and_goals`	augmented_matrix, elimination_and_rref, gauss_jordan_elimination, linear_system, reduced_row_echelon_form	—
`roundoff_and_partial_pivoting`	elimination_and_rref, gauss_jordan_elimination, linear_system, pivot, pivoting	—
`ill_conditioning`	ill_conditioning, linear_system, linear_systems_core	ill_conditioning
`polynomial_interpolation_example`	elimination_and_rref, gauss_jordan_elimination, ill_conditioning, linear_system, linear_systems_core, pivoting	—

Day 2 — More Systems of Linear Equations (`day_2.qmd`)

Idea	Uses	Introduces
`rref_free_bound_and_consistency`	augmented_matrix, basic_column, consistent_system, elimination_and_rref, free_and_bound_variables, homogeneous_system, linear_system, linear_systems_core, nullity, rank, rank_and_nullity, reduced_row_echelon_form	free_and_bound_variables, basic_column, consistent_system, rank, nullity, homogeneous_system
`pagerank_tutorial`	distribution_vector, dynamical_and_markov, markov_chain, stationary_vector, stochastic_matrix, transition_matrix	—
`diffusion_discretization`	linear_system	—
`reaction_diffusion_biology`	linear_system	—

Day 3 — Ch2 Lecture 1 (`day_3.qmd`)

Idea	Uses	Introduces
`diffusion_and_boundary_conditions`	linear_system	—
`reaction_diffusion_solving`	linear_system	—
`matrix_arithmetic_review`	linear_combination, linear_system, linear_map, matrix_arithmetic	linear_combination, linear_map
`transformations_scaling_rotation`	linear_map, matrix_arithmetic	—

Day 4 — Ch2 Lecture 2 (`day_4.qmd`)

Idea	Uses	Introduces
`dynamical_systems_and_markov`	discrete_dynamical_system, distribution_vector, dynamical_and_markov, markov_chain, stationary_vector, stochastic_matrix, transition_matrix	discrete_dynamical_system, transition_matrix, stationary_vector, distribution_vector, stochastic_matrix, markov_chain
`graphs_and_adjacency`	adjacency_matrix, directed_graph, graph, graphs	graph, directed_graph, adjacency_matrix

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Day 5 — Ch2 Lecture 3 (`day_5.qmd`)

Idea	Uses	Introduces
`difference_equations_intro`	linear_system	—
`digital_filters_examples`	discrete_filters, highpass_filter, lowpass_filter	lowpass_filter, highpass_filter
`inverse_matrices_and_algorithm`	elementary_matrix, inverse_matrix, inverses_and_factorization, superaugmented_matrix	inverse_matrix, elementary_matrix, superaugmented_matrix
`pagerank_as_markov`	dynamical_and_markov, markov_chain, stationary_vector, transition_matrix	—

Day 6 — Ch2 Lecture 4 (`day_6.qmd`)

Idea	Uses	Introduces
`mcmc_and_gibbs_sampling`	markov_chain	—
`rbm_training_and_gibbs`	markov_chain	—
`lu_factorization_saving_work`	inverses_and_factorization, linear_system, lu_factorization	lu_factorization
`plu_factorization`	inverses_and_factorization, lu_factorization, permutation_matrix, superaugmented_matrix	permutation_matrix

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Day 7 — Ch2 Lecture 5 (`day_7.qmd`)

Idea	Uses	Introduces
`block_multiplication`	block_multiplication, matrix_arithmetic	block_multiplication
`transpose_and_symmetric`	matrix_arithmetic, transpose	transpose
`inner_and_outer_products`	matrix_arithmetic, transpose	—
`quadratic_forms`	transpose	—
`determinants_and_invertibility`	determinant, inverse_matrix, inverses_and_factorization	determinant

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Day 8 — Ch3 Lecture 1 (`day_8.qmd`)

Idea	Uses	Introduces
`linear_independence_and_invertibility`	independence_and_span, inverse_matrix, linear_independence	linear_independence
`vector_spaces_and_basis`	basis, fundamental_subspaces, independence_and_span, span, subspace	—
`spanning_sets_and_standard_basis`	basis, independence_and_span, span	span, basis
`fundamental_subspaces`	column_space, fundamental_subspaces, left_null_space, null_space, row_space, subspace	subspace, column_space, row_space, null_space
`applications_markov_and_consistency`	consistent_system, dynamical_and_markov, markov_chain, stationary_vector, transition_matrix	—

Day 9 — Ch4 Lecture 1 (`day_9.qmd`)

Idea	Uses	Introduces
`column_space_geometry`	column_space, fundamental_subspaces	—
`planes_and_projections`	column_space, least_squares_and_projection, projection	left_null_space, projection
`least_squares_and_normal_equations`	least_squares, least_squares_and_projection, normal_equations, projection	least_squares, normal_equations

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Day 10 — Ch4 Lecture 2 (`day_10.qmd`)

Idea	Uses	Introduces
`normal_equations_multiple_predictors`	least_squares, least_squares_and_projection, normal_equations, projection	—
`orthogonal_and_orthonormal_sets`	orthogonal_and_qr, orthogonal_set, orthonormal_set	orthogonal_set, orthonormal_set, orthogonal_matrix, unitary_matrix
`gram_schmidt_and_orthonormal_basis`	gram_schmidt, orthogonal_and_qr, orthonormal_set	gram_schmidt
`qr_factorization_and_least_squares`	least_squares, orthogonal_and_qr, qr_factorization	qr_factorization
`haar_wavelet_and_compression`	orthonormal_set, qr_factorization	—

Day 11a — Ch5 Lecture 1 (`day_11a.qmd`)

Idea	Uses	Introduces
`eigenvalues_eigenspaces_algorithm`	diagonalization, eigenvalue, eigenvector, eigenspace, eigen-value_eigenvector	eigenvalue, eigenvector, eigenspace
`diagonalization_and_similarity`	diagonalization, eigenvalue, eigenvector, eigen-value_eigenvector	diagonalization
`spectral_radius_and_dynamics`	dominant_eigenvalue, eigenvalue, eigen-value_eigenvector, spectral_radius	spectral_radius, dominant_eigenvalue

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Day 12a — Ch5 Lecture 2 (`day_12a.qmd`)

Idea	Uses	Introduces
`symmetric_matrices_spectral_theorem`	diagonalization, eigenvalue, eigenvector, eigenvalue_eigenvector, orthogonal_set	—
`diagonalizing_quadratic_forms`	diagonalization, eigenvalue, eigenvector, eigenvalue_eigenvector	—
`coupled_springs_diagonalization`	diagonalization, eigenvalue, eigenvector, eigenvalue_eigenvector	—
`singular_values_and_svd_theorem`	eigenvalue, eigenvector, singular_value, singular_value_decomposition, svd, svd_core	singular_value, right_singular_vector, left_singular_vector, singular_value_decomposition

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Day 12b — Ch5 Lecture 3 (`day_12b.qmd`)

Idea	Uses	Introduces
`svd_from_diagonalization_intuition`	diagonalization, eigenvalue, eigenvector, singular_value_decomposition, svd	—
`pseudoinverse_and_least_squares`	least_squares, least_squares_and_projection, singular_value_decomposition, svd	—

Idea	Uses	Introduces
`data_matrices_and_covariance`	least_squares, singular_value_decomposition, svd	—

SVD Interlude (`svd_interlude.qmd`)

Idea	Uses	Introduces
`input_output_orthogonal_directions`	column_space, left_singular_vector, right_singular_vector, singular_value, singular_value_decomposition, svd_core	—
`from_rank_to_svd_matrix_form`	rank, singular_value_decomposition, svd, svd_core	—

Day 14 — Ch5 Lecture 4 (`day_14.qmd`)

Idea	Uses	Introduces
`svd_on_tabular_data`	left_singular_vector, right_singular_vector, singular_value, singular_value_decomposition, svd, svd_core	—
`svd_on_images_and_pca`	pca, principal_component, singular_value_decomposition, svd, svd_core	principal_component

Idea	Uses	Introduces
`pca_high_dimensional_data`	pca, principal_component, singular_value_decomposition, svd	—

Day 15 — Ch5 Lecture 5 (`day_15.qmd`)

Idea	Uses	Introduces
`high_dimensional_data_and_projection`	projection	—
`first_principal_component_definition`	eigenvalue, eigenvector, pca, principal_component, right_singular_vector, singular_value	—
`shopping_baskets_exploration`	pca, principal_component	—
`interpreting_principal_components`	pca, principal_component	—

Day 18a — Ch6 Fourier (`day_18a.qmd`)

Idea	Uses	Introduces
`discrete_filters_dtft_and_gain`	discrete_filter, discrete_filters, discrete_time_fourier_transform, filter_gain, finite_impulse_response_filter, phase_rotation	discrete_filter, finite_impulse_response_filter, discrete_time_fourier_transform, filter_gain, phase_rotation

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Day 18 — Review (`day_18_review.qmd`)

Idea	Uses	Introduces
`review_linear_systems`	elimination_and_rref, linear_system, linear_systems_core, reduced_row_echelon_form	—
`review_matrices`	linear_combination, linear_map, matrix_arithmetic	—
`review_vector_spaces`	basis, column_space, fundamental_subspaces, independence_and_span, linear_independence, null_space, span, subspace	—
`review_orthogonality`	least_squares, least_squares_and_projection, normal_equations, orthogonal_and_qr, projection, qr_factorization	—
`review_eigenvalues_svd_pca`	eigenvalue, eigenvector, eigenvalue_eigenvector, pca, principal_component, singular_value_decomposition, svd	—
`review_fourier`	discrete_filters, discrete_time_fourier_transform, filter_gain	—

ChNone — Fourier (standalone) (`ChNone.qmd`)

Idea	Uses	Introduces
`fourier_series_and_orthogonality`	orthogonal_set, orthonormal_set	—
`discrete_fourier_series_and_dft`	discrete_fourier_transform, discrete_filters	—
`dft_filtering_and_convolution`	discrete_filter, discrete_filters, filter_gain	—

Test (`test.qmd`)

Idea	Uses	Introduces
`test_slide`	linear_system	—

Generated from `@idea`, `@uses`, and `@introduces` tags in lecture `.qmd` files. Concept and group ids match `course_concepts.yaml`.